

Sri Lanka Institute of Information Technology



Streetify

Web-Based Transportation System.

Lab Report 04: Activity Modeling & Scenarios

Group ID : 2026-Y2-S1-KU-24 (Group 24)

IT25102193	Weerasingha W.A.M.B.
IT25102207	Dharmakeerthi W.A.C.B.
IT25102208	Nayanamina A.R.L.P.
IT25102225	Edirisinghe E.A.R.N.D.
IT25102240	Rammandalagedara R.V.S.
IT25102241	Senaka K.A.T.

Group Members & Assigned Modules:

IT number	Name	Functions / Assigned Modules
IT25102208	Nayanamina A.R.L.P.	User Account Management
IT25102207	Dharmakeerthi W.A.C.B.	Ride Booking & Management
IT25102241	Senaka K.A.T.	Driver Trip Request Management
IT25102225	Edirisinghe E.A.R.N.D.	Payment & Receipt Management
IT25102193	Weerasingha W.A.M.B.	Review, Rating & Feedback Management
IT25102240	Rammandalagedara R.V.S.	Administration & System Control

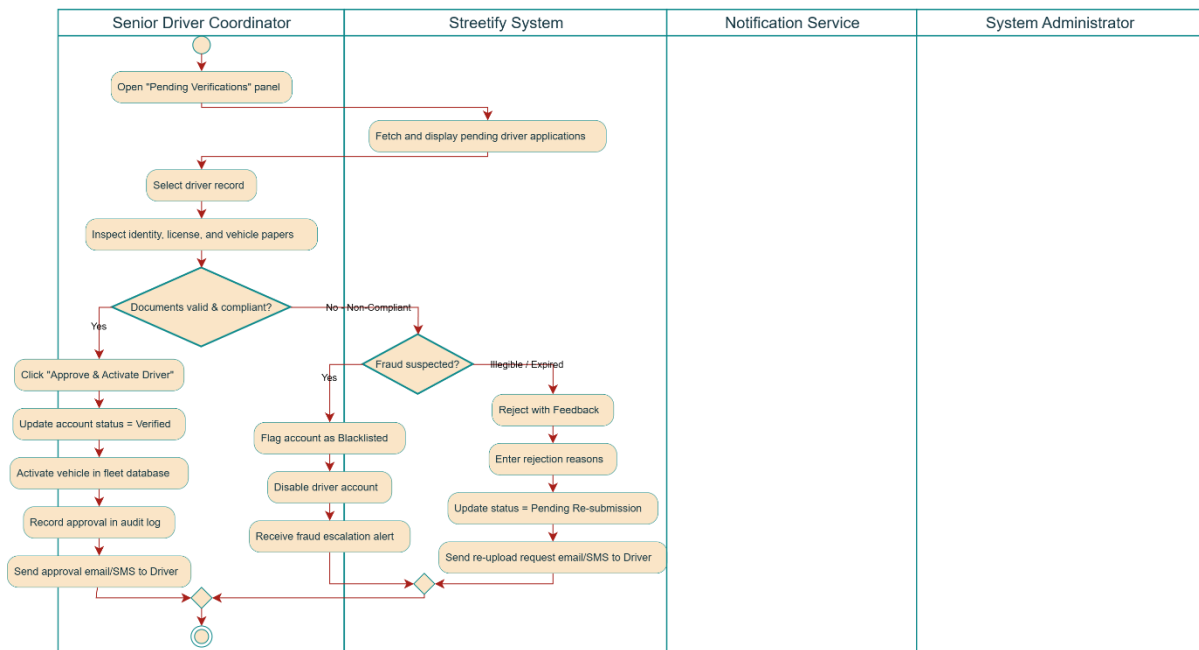
Member 1: IT25102208 – Nayanamina A.R.L.P.

✓ Functional Module: User Account Management

1.1 Use Case Scenario: Verify Driver Registration Documents

Field	Description
Use Case ID	UC-UAM-01
Use Case Name	Verify Driver Registration Documents
Primary Actor	Senior Driver Coordinator (Mr. Kasun Jayawardena)
Supporting Actor	Driver, Notification Service
Description	Allows the Senior Driver Coordinator to review submitted identity proof, driving license, and vehicle registration documents to certify a driver for commercial service on Streetify.
Pre-Conditions	1. Driver has registered and uploaded legible copies of license and insurance. 2. Driver Coordinator is logged into the Streetify Admin Portal with valid credentials.
Post-Conditions	Driver account status is transitioned to Active/Approved, vehicle is registered into the active fleet, and an automated approval email/SMS notification is dispatched.
Main Success Scenario (Basic Flow)	1. Coordinator navigates to the "Pending Driver Verifications" panel. 2. System displays a queued list of unverified driver accounts. 3. Coordinator selects a specific driver application record. 4. System renders the driver's profile details, driving license image, revenue license, and vehicle inspection papers. 5. Coordinator verifies the validity dates, name matching, and regulatory compliance of all documents. 6. Coordinator clicks the "Approve & Activate Driver" button. 7. System updates the driver's database record to Verified status. 8. System triggers the Notification Service to dispatch an onboarding welcome alert to the driver.
Extensions (Alternative Flows)	4a. Illegible or Expired Documents Provided: 1. Coordinator selects "Reject with Feedback" and selects reason (e.g., "Expired Revenue License"). 2. System sets driver status to Pending Re-submission and emails instructions to re-upload. 5a. Suspected Fraudulent Documents: 1. Coordinator flags the account as Blacklisted and escalates to System Administrator.

1.2 Activity Diagram: Verify Driver Registration Documents



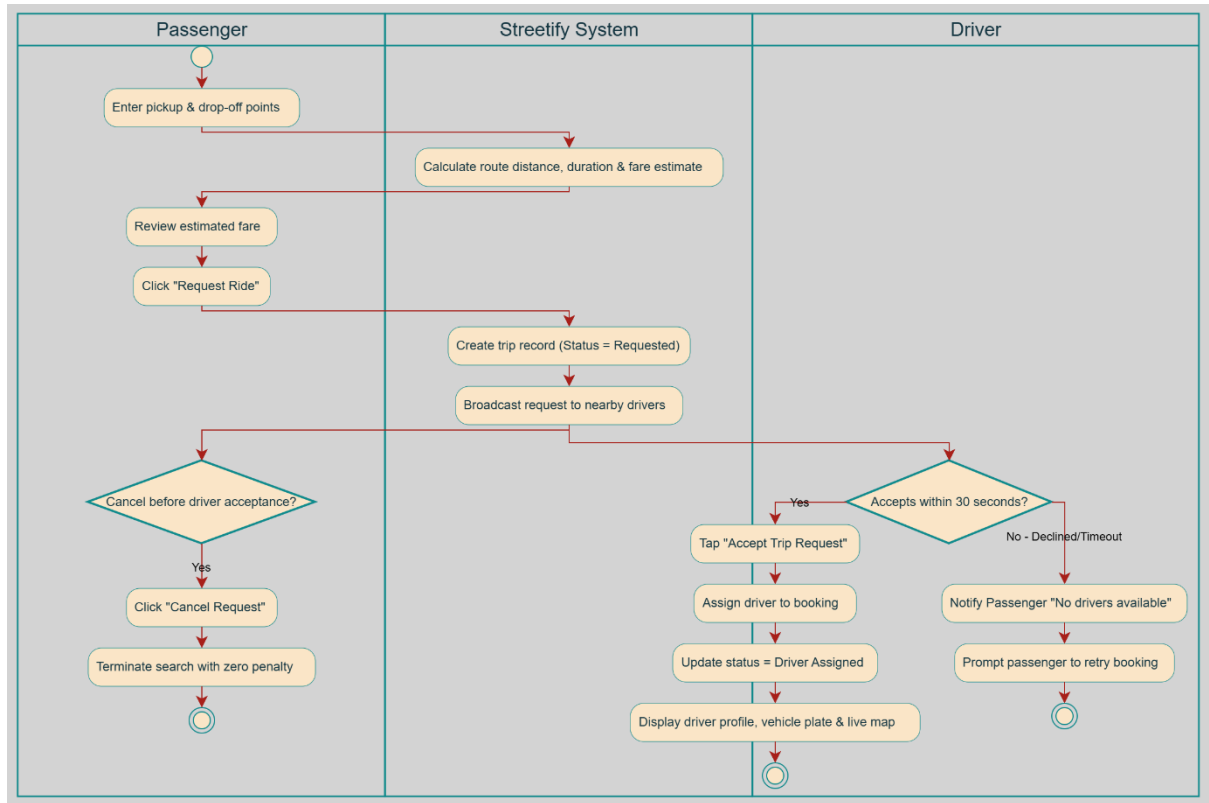
Member 2: IT25102207 – Dharmakeerthi W.A.C.B.

✓ Functional Module: Ride Booking & Management

2.1 Use Case Scenario: Request a Ride and Match Driver

Field	Description
Use Case ID	UC-RBM-01
Use Case Name	Request a Ride and Match Driver
Primary Actor	Passenger
Supporting Actor	Driver, Notification Service
Description	Enables a passenger to specify pickup and destination locations, review calculated fare estimates, and initiate a ride booking request that matches with nearby available drivers.
Pre-Conditions	1. Passenger is authenticated and logged in. 2. Passenger device has GPS/location services enabled on the browser.
Post-Conditions	A trip request entity is created in Requested status, matched with an accepting driver, and live tracking is initialized.
Main Success Scenario (Basic Flow)	1. Passenger opens the Streetify Ride screen and enters pickup and drop-off points. 2. System calls map route calculation API and computes distance, duration, and estimated fare. 3. Passenger reviews estimated fare and confirms booking by clicking "Request Ride". 4. System broadcasts the trip request to active drivers within the municipal radius. 5. A nearby driver accepts the trip request within the 30-second window. 6. System assigns the driver to the booking and updates ride status to Driver Assigned. 7. System displays driver name, vehicle model, license plate number, rating, and live location map to the passenger.
Extensions (Alternative Flows)	4a. No Drivers Available in the Vicinity: 1. System fails to find an available driver after 60 seconds. 2. System alerts passenger: "No drivers available currently. Please try again shortly." 5a. Passenger Cancels Before Driver Acceptance: 1. Passenger clicks "Cancel Request". 2. System cancels the broadcast request immediately without penalty fees.

2.2 Activity Diagram: Request a Ride and Match Driver



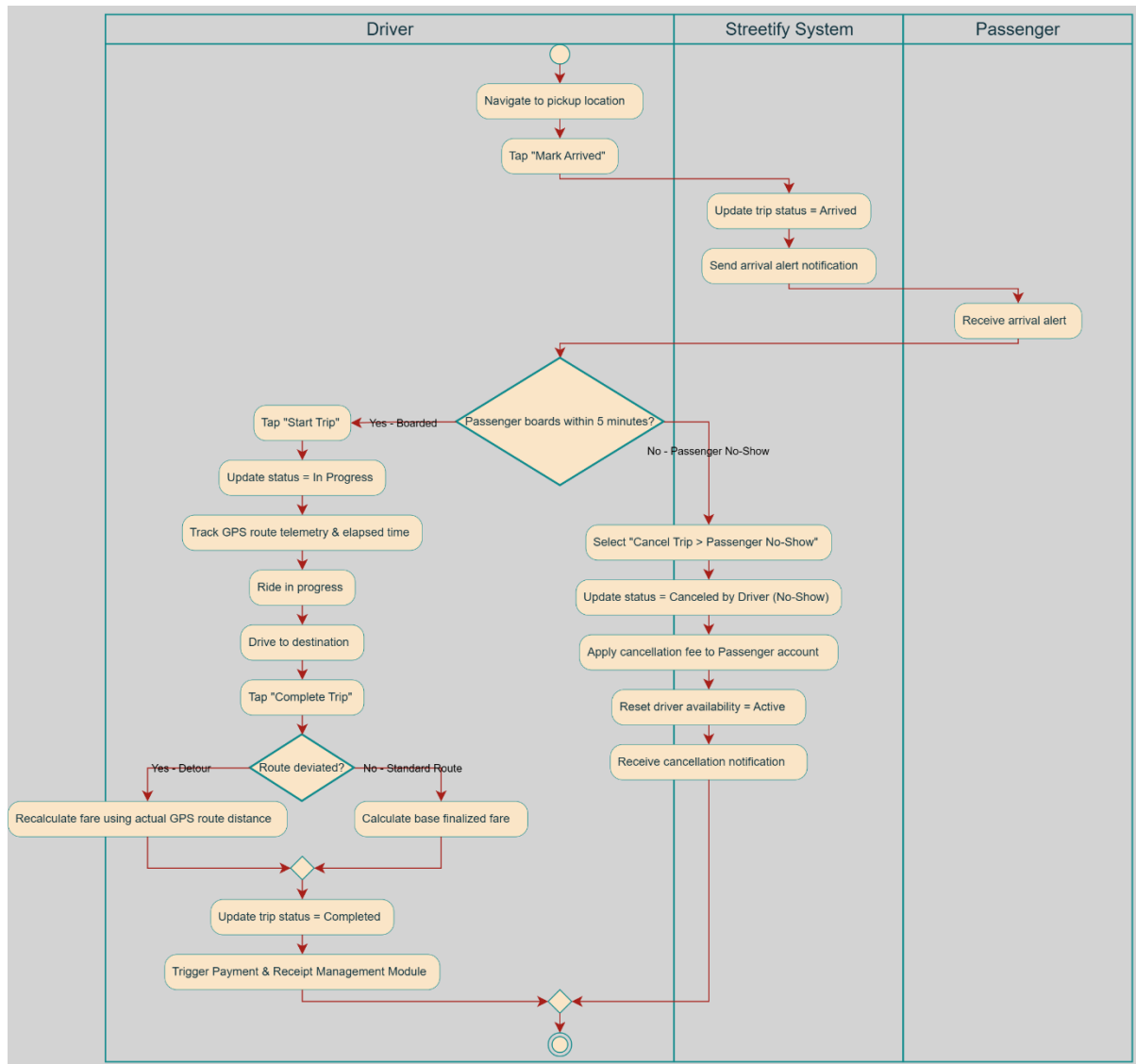
Member 3: IT25102241 – Senaka K.A.T.

✓ Functional Module: Driver Trip Request Management

3.1 Use Case Scenario: Update Trip Progress Status

Field	Description
Use Case ID	UC-DTM-01
Use Case Name	Update Trip Progress Status
Primary Actor	Driver
Supporting Actor	Passenger, Payment Module
Description	Enables an assigned driver to update the sequential milestones of an active ride from arriving at the pickup point through trip completion.
Pre-Conditions	1. Driver has an accepted and active trip assigned. 2. Driver is logged in on a mobile/desktop web browser.
Post-Conditions	Trip record status changes to Completed, trip duration and distance are locked, and the payment workflow is initiated.
Main Success Scenario (Basic Flow)	1. Driver reaches the passenger's pickup location and taps "Mark Arrived". 2. System updates trip status to Arrived and sends arrival push/SMS alert to passenger. 3. Passenger boards vehicle; driver clicks "Start Trip". 4. System transitions state to In Progress and starts tracking real-time transit telemetry. 5. Upon reaching the destination, driver clicks "Complete Trip". 6. System calculates final actual fare based on exact route distance/time. 7. System changes trip state to Completed and redirects the driver to the fare summary and collection screen.
Extensions (Alternative Flows)	1a. Driver Cancels after Arrival (Passenger No-Show): 1. Driver waits for >5 minutes and selects "Cancel Ride > Passenger Did Not Show". 2. System logs cancellation and applies standard no-show policy fee to passenger. 5a. Route Detour Encountered: 1. System dynamically adjusts final distance and updates total fare calculation prior to completion.

3.2 Activity Diagram: Update Trip Progress Status



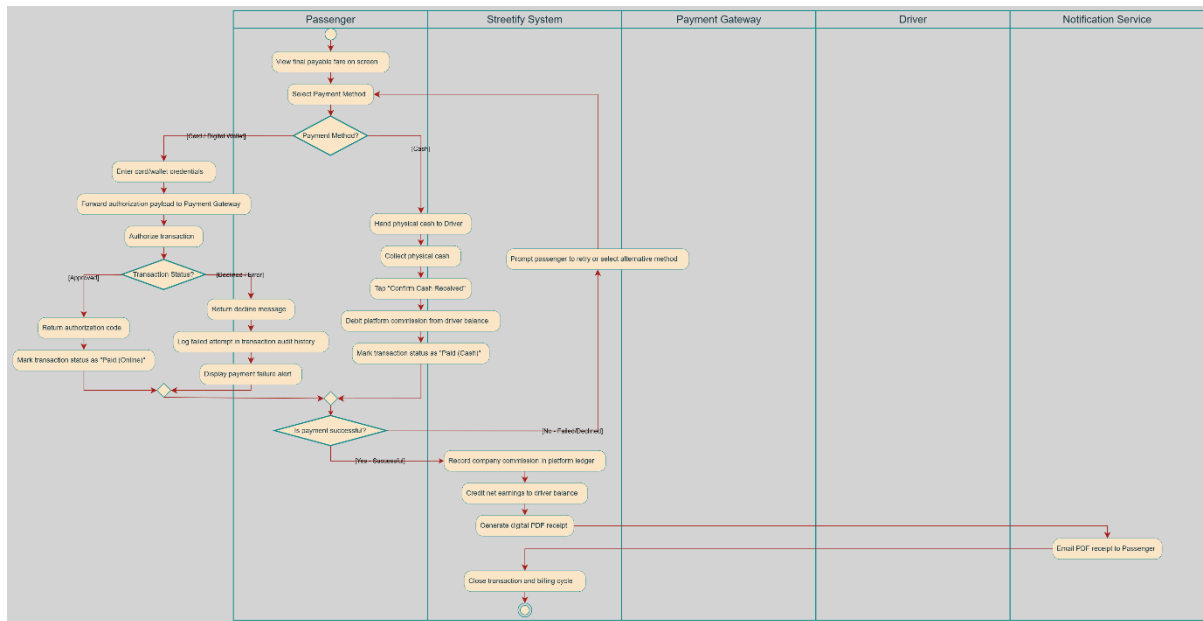
Member 4: IT25102225 – Edirisinghe E.A.R.N.D.

✓ Functional Module: Payment & Receipt Management

4.1 Use Case Scenario: Process Multi-Channel Payment & Generate Receipt

Field	Description
Use Case ID	UC-PRM-01
Use Case Name	Process Multi-Channel Payment & Generate Receipt
Primary Actor	Passenger
Supporting Actor	Payment Gateway, Driver, Finance Manager, Notification Service
Description	Executes final payment processing via Credit/Debit Card, Digital Wallet, or Cash, records platform commissions, and generates an automated electronic receipt.
Pre-Conditions	Trip status has been marked as Completed by the driver with a finalized bill amount.
Post-Conditions	Payment record is marked Paid, commission split is credited to company ledger, driver earnings balance is updated, and a PDF receipt is dispatched.
Main Success Scenario (Basic Flow)	1. System displays final payable fare on passenger dashboard upon trip completion. 2. Passenger selects preferred payment method (Credit/Debit Card or Digital Wallet). 3. Passenger submits card/wallet payment request. 4. System communicates with external Payment Gateway for tokenized payment capture. 5. Payment Gateway returns Transaction Approved with transaction reference code. 6. System deducts company commission rate and credits remainder to driver's balance. 7. System generates an electronic invoice/receipt PDF with trip details, fare breakdown, and tax info. 8. System delivers receipt on-screen and automatically emails copy via Notification Service.
Extensions (Alternative Flows)	2a. Passenger Chooses Cash Payment: 1. Driver collects cash physically and clicks "Confirm Cash Payment Received". 2. System records cash collection, adjusts driver's digital platform commission debt, and issues digital receipt. 5a. Card Transaction Declined / Gateway Error: 1. Payment Gateway returns Insufficient Funds / Declined. 2. System prompts passenger to retry or switch to alternative card/cash. 3. Failed attempt is logged in finance transaction ledger.

4.2 Activity Diagram: Process Multi-Channel Payment & Generate Receipt



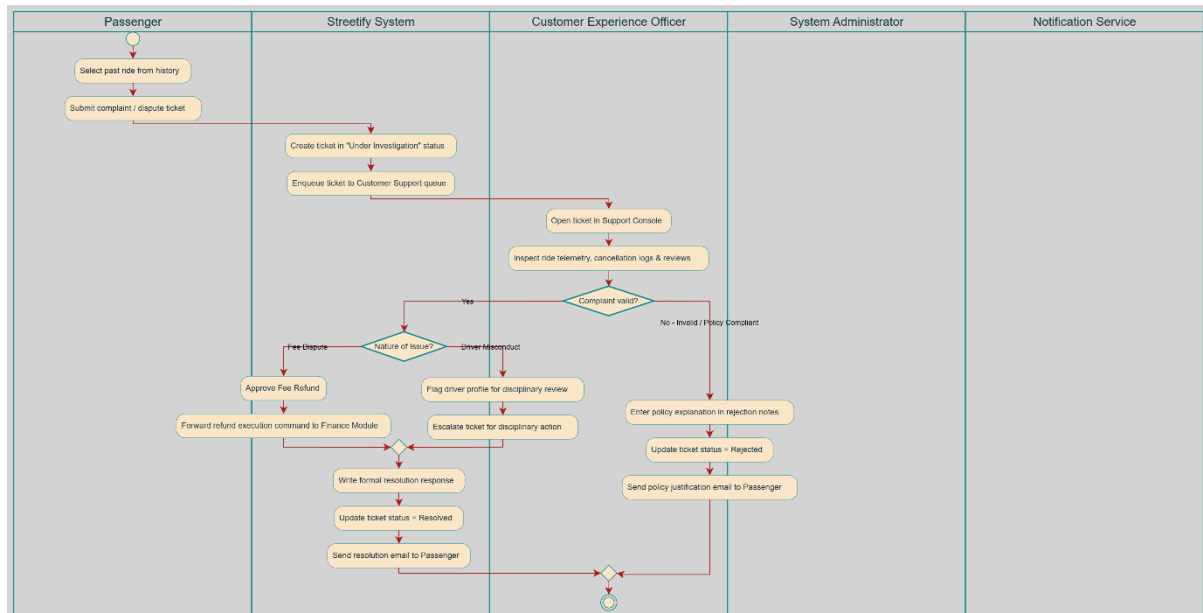
Member 5: IT25102193 – Weerasingha W.A.M.B.

✓ Functional Module: Review, Rating & Feedback Management

5.1 Use Case Scenario: Submit and Investigate Customer Complaint

Field	Description
Use Case ID	UC-RFM-01
Use Case Name	Submit and Investigate Customer Complaint
Primary Actor	Passenger, Customer Experience Officer (Ms. Tharushi Fernando)
Supporting Actor	System Administrator, Notification Service
Description	Allows a passenger to file a complaint or cancellation fee dispute for a recent trip, and provides support staff tools to investigate and formally resolve the issue.
Pre-Conditions	1. Passenger has completed or canceled a trip within the last 7 days. 2. Customer Experience Officer is logged into the Customer Support Console.
Post-Conditions	Complaint ticket is updated to Resolved, resolution response is emailed to the user, and corrective action (e.g., refund trigger or driver warning) is logged.
Main Success Scenario (Basic Flow)	1. Passenger selects past ride from history and clicks "Report Issue / Dispute Charge". 2. Passenger fills complaint category, text explanation, and submits support ticket. 3. System creates a ticket with Under Investigation status and alerts the support queue. 4. Customer Experience Officer opens the complaint ticket in the support dashboard. 5. Officer inspects linked telemetry logs, GPS route deviations, cancellation time, and driver feedback history. 6. Officer approves passenger dispute as valid unfair cancellation fee. 7. Officer writes formal resolution response and selects "Issue Fee Refund" action. 8. System forwards refund command to Finance Module and dispatches official resolution email to passenger.
Extensions (Alternative Flows)	6a. Officer Finds Complaint Invalid / Policy Compliant: 1. Officer enters explanation detailing why policy justified the fee. 2. System closes ticket as Rejected and notifies passenger with policy documentation. 6b. Complaint Involves Driver Misconduct: 1. Officer flags driver profile and escalates ticket to Senior Driver Coordinator / Admin for disciplinary warning.

5.2 Activity Diagram: Submit and Investigate Customer Complaint



Member 6: IT25102240 – Rammandalagedara R.V.S.

✓ Functional Module: Administration & System Control

6.1 Use Case Scenario: Suspend Policy-Violating Account & Enforce Governance

Field	Description
Use Case ID	UC-ASC-01
Use Case Name	Suspend Policy-Violating Account & Enforce Governance
Primary Actor	System Administrator (Mr. Ravindu Senanayake)
Supporting Actor	Senior Driver Coordinator, Notification Service
Description	Enables the System Administrator to monitor security logs, inspect accounts committing severe policy violations (e.g., fraudulent activities, excessive safety complaints), and enforce account suspensions or deactivations.
Pre-Conditions	1. System Administrator is authenticated with highest-tier RBAC credentials. 2. Flagged user or driver account has logged severe infractions or safety reports.
Post-Conditions	Target account status is set to Suspended, active session tokens are revoked, and login capability is blocked.
Main Success Scenario (Basic Flow)	1. Administrator accesses the "Security & Audit Governance Dashboard". 2. System displays platform audit logs, flagged incident reports, and accounts exceeding cancellation thresholds. 3. Administrator selects a flagged driver or passenger account for detailed audit. 4. System renders infraction logs, customer complaint tickets, and historical ride telemetry. 5. Administrator confirms repeated violation of SwiftRide safety regulations. 6. Administrator clicks "Suspend Account", selects duration (e.g., 14-day ban or Permanent Deactivation), and enters audit notes. 7. System revokes active JWT sessions, terminates any live ride availability, and sets status to Suspended. 8. System records administrative audit log and triggers automated suspension notice email to the account owner.
Extensions (Alternative Flows)	3a. System Administrator Reviews Re-activation Appeal: 1. Admin reviews submitted appeal documentation from a suspended user. 2. Admin selects "Reactivate Account" and restores system permissions.

6.2 Activity Diagram: Suspend Policy-Violating Account & Enforce Governance

